



FORCE SENSOR CHRONOJUMP

ISOMETRIC EXERCISES GUIDE

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Association Chronojump

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This guide is intended to support the practitioner in using the force sensor with a set of isometric exercises. The guide develops the creation of exercises and specifies the execution and presents supporting images.

For complete information on the use of Chronojump refer to the Chronojump manual on the website or in the software in the help section.

First of all, thanks to Alex Rafa and Javi Celeiro for their photography work, to all the Chronojump team for the help in the realization of this guide, and to the INEFC of Barcelona for the use of their facilities.

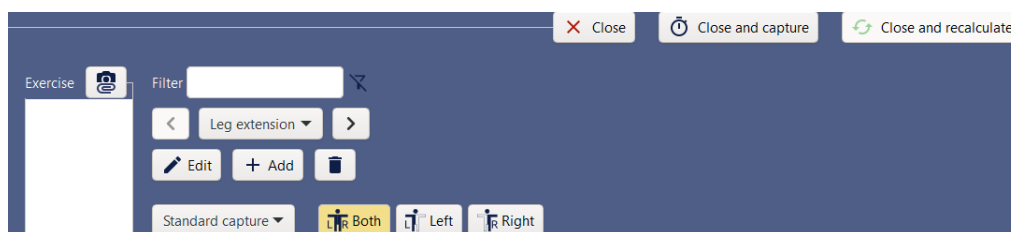
EXERCISE CREATION GUIDE IN CHRONOJUMP

For the correct performance of all these force tests it is essential to have the Force Sensor Kit and a computer with the latest version of Chronojump installed. In addition, the Chronojump program must be used in Isometric mode having created and added the exercise to be performed. When creating exercises it must be taken into account that if they are performed standing and against gravity there is an attempt of displacement of the body mass and, if they are performed with a barbell, there is also an attempt of displacement and its weight must be known.

To create a new exercise, go to the "Capture" box and click on "Exercise".



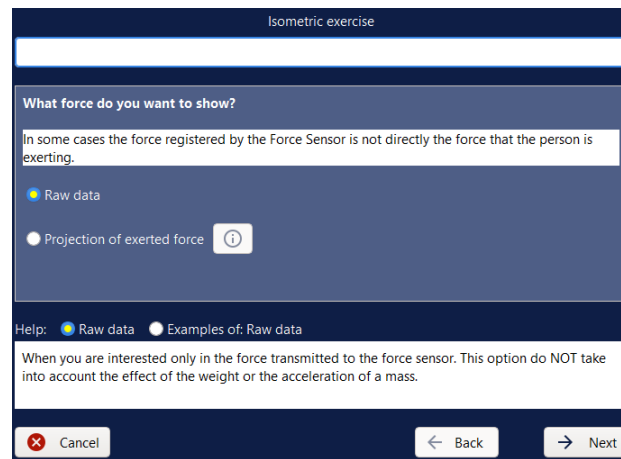
In the pop-up tab select "Add".



A window will open where you name the exercise, and two options from which you have to select one:

- "Raw data": when you are interested only in the force transmitted to the force sensor.
- "Projection of the force exerted": when you want the result of all the forces exerted by the athlete. Chronojump needs to know the mass involved (usually the mass of the person) and the angle. This option allows you to take into account the effect of the weight or acceleration of a mass. The projection of the force exerted is calculated as the sum of the forces projected in the direction defined by the exercise.

If when performing an exercise there are doubts about which option is appropriate, it is best to select "Raw data".



Isometric exercise

What force do you want to show?

In some cases the force registered by the Force Sensor is not directly the force that the person is exerting.

☒ Raw data

☐ Projection of exerted force ⓘ

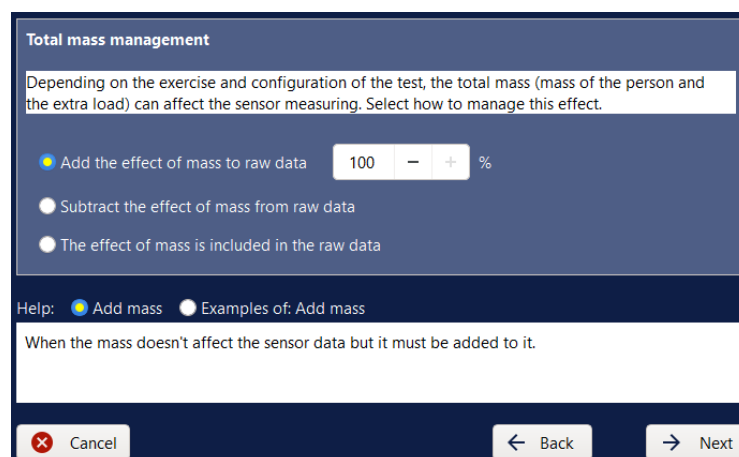
Help: ☒ Raw data ☐ Examples of: Raw data

When you are interested only in the force transmitted to the force sensor. This option do NOT take into account the effect of the weight or the acceleration of a mass.

Cancel Back Next

If "Raw data" has been selected, clicking "Next" will change the window to "Detect repetitions". If "Projection of exerted force" has been selected, clicking "Next" will change the window to "Total mass management", a step before "Detect repetitions". In this window there are three options:

- "Add the effect of the mass to raw data": when the mass does not affect the sensor data but must be added to it. With this option you also have to specify the percentage of the body mass to be added.
- "Subtract the effect of the mass from raw data": in some cases the weight of the mass is supported by the sensor but is not the force that the subject is exerting. In this case, the sensor will be tared before starting each test. A clear example of this case is the knee flexion exercise (hamstring), where the leg is suspended and held by the strap followed by the force sensor, so that the sensor detects the leg weight as a force exerted.
- "Mass effect of mass is included in the raw data": in some cases the weight is transmitted to the sensor and is also supported by the measured limb. If the mass effect is not significant use this option. This setting is the same as for the raw data.



Total mass management

Depending on the exercise and configuration of the test, the total mass (mass of the person and the extra load) can affect the sensor measuring. Select how to manage this effect.

☒ Add the effect of mass to raw data 100 - + %

☐ Subtract the effect of mass from raw data

☐ The effect of mass is included in the raw data

Help: ☒ Add mass ☐ Examples of: Add mass

When the mass doesn't affect the sensor data but it must be added to it.

Cancel Back Next

Once you have selected one of the three options, clicking "Next" will take you to "Detect repetitions". In this case there are two options:

- "Use value on preferences to detect repetitions": if the user changes the values in the preferences, these values will change automatically.
- "Use custom value": these values will be used to detect repetitions. With this option you have to specify custom values for this exercise.

The screenshot shows a dialog box titled "Detect repetitions". It has two radio button options: "Use value on preferences to detect repetitions" (which is selected) and "Use custom value". Below the first option, the text "Minimal range of movement Eccentric 100 N Concentric 100 N" is displayed. Below the second option, there are input fields for "Minimal force", "Eccentric" (with a value of 100), and "Concentric" (with a value of 100), each followed by a minus/plus button and the unit "N". A "Help" section at the bottom left shows the selected option and a text box stating: "If user changes values on preferences, these values will automatically change." At the bottom right are three buttons: "Cancel", "Back", and "Next".

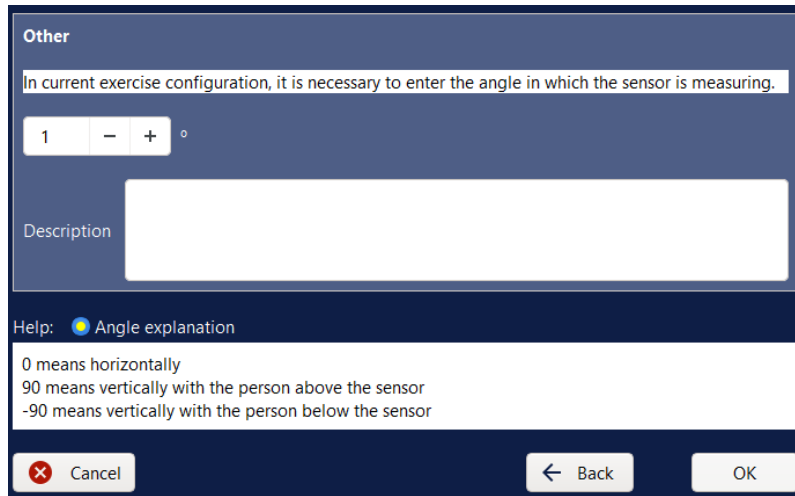
The next screen is the "Show repetitions" screen, and contains two options:

- "Show only the phase in which the force increases": the detected repetitions will only be shown in the phase in which the force increases.
- "Show both phases": the detected repetitions will be displayed both in the phase in which the force increases and when it decreases.

If this option is selected, two more options will appear: "Together" and "Separated".

The screenshot shows a dialog box titled "Show repetitions". It has two radio button options: "Show only the phase in which the force increases" and "Show both phases" (which is selected). Below the second option, there are two more radio button options: "Together" (which is selected) and "Separated". A "Help" section at the bottom left shows the selected option and a text box stating: "Detected repetitions will show only the phase in which the force increases or also where decreases." At the bottom right are three buttons: "Cancel", "Back", and "Next".

Finally, in the last screen "Other", enter the angle at which the sensor is measuring. Note that 0 means horizontally, 90 means vertically with the person above the sensor, and -90 means vertically with the person below the sensor.



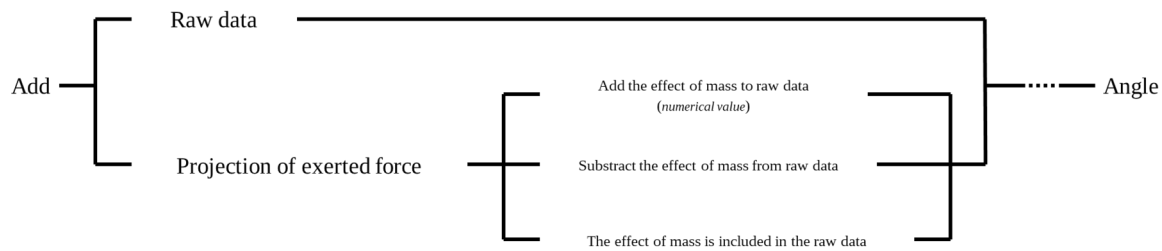
The screenshot shows a dark-themed window titled "Other". At the top, a text box contains the instruction: "In current exercise configuration, it is necessary to enter the angle in which the sensor is measuring." Below this is a numeric input field with the value "1", flanked by minus and plus buttons, and a degree symbol. Underneath is a "Description" label followed by a large empty text area. At the bottom, there is a "Help:" section with a blue lightbulb icon and the text "Angle explanation". This section contains three lines of text: "0 means horizontally", "90 means vertically with the person above the sensor", and "-90 means vertically with the person below the sensor". At the very bottom are three buttons: "Cancel" (with a red X icon), "Back" (with a left arrow icon), and "OK".

To finish creating the exercise click "Ok", the pop-up window will close and you will return to the window where you started. Before closing the tab for calculating you should select the exercise literal.



EXERCISE BATTERY

Below is the battery of exercises. In each exercise the first thing you will find is how to create the exercise following the scheme below:



In many exercises a ground anchor is required. If one is not available, a bar can be attached to the strap that would be anchored to the floor and, with an assistant to step on it or by stepping on it yourself, an alternative anchor is already available (example in the mid-thigh pull).

In addition, as what is sought with these exercises is isometric strength, it is also important that the straps are taut to avoid peaks of elastic force.

1. Upper trunk

1.1. Neck

1.1.1. Extension

Raw data → 90°

To properly assess the strength of the extensor muscles of the neck, the following is needed:

- A bench.
- Two straps, one of which must be padded so that the athlete can perform the test without any discomfort.
- A floor anchor.

The procedure to perform the strength test is as follows:

1. The unpadded strap is attached to the floor, followed by the force sensor and the other strap, and the bench is placed close to the anchor.
2. The subject lies face down on the bench so that the head and neck protrude.
3. The subject places the padded strap around the head and the straps are tightened.
4. The exercise can now be performed (Figure 1).



Figure 1: Neck extension.

1.1.2. Flexion

Raw data → 90°

The material and procedure to perform this exercise is exactly the same as for the neck extension, only in this one, instead of lying face down, the subject has to lie face up (Figure 2).



Figure 2: Neck flexion.

1.2. Shoulders

1.2.1. Extension

Raw data → 0°

To properly assess shoulder extension strength, the following equipment is needed:

- A bench.
- Two straps, one of which has to be padded as in the image (Figure 3), or has to have a handle for gripping.
- An anchor point at a certain height that allows the strap to be attached.

The procedure for this test is as follows:

1. The unpadded strap is attached to the anchor point and placed on bench as shown in the picture.
2. The subject lies face up on the bench, so that the arm is fully extended (if necessary, the height of the anchor point or the bench should be modified) and at ninety degrees to the body and the bench.
3. The subject grasps the strap and can now perform the exercise.



Figure 3: Shoulder extension.

1.2.2. Flexion

Raw data → 90°

To properly assess the strength of shoulder extension, the following equipment is needed:

- A bench, stool or any seat on which neither the arms nor the back are supported.
- Two straps.
- A floor anchor.
- A light bar.

The procedure for this test is as follows:

1. One strap is anchored to the anchor point and the other to the light bar.
2. The seat is placed close to the anchor point and the subject sits so that when grasping the bar the arms are fully extended and at less than ninety degrees to the trunk, and so that the bar and the straps are just above the anchor point, perpendicular to the floor.
3. The subject grasps the bar and can now perform the exercise (Figure 4).



Figure 4: Shoulder flexion.

1.2.3. Adduction

Raw data → 0°

To properly assess the adduction strength of the shoulders, the following equipment is needed:

- A bench.
- Two straps, one of which has to be padded as in the image (Figure 5 and 6), or has to have a handle for gripping.
- An anchor point at a certain height that allows the strap to be attached.

The procedure for this test is as follows:

1. The unpadded strap is attached to the anchor point and the bench is placed parallel to the surface where the anchor point is located.
2. The subject lies face up on the bench so that the arm is fully extended (if necessary, the height of the anchor point or the bench should be modified) and at ninety degrees to the body and the bench.
3. The subject grasps the strap and can now perform the exercise.



Figure 5: Shoulder adduction.



Figure 6: Shoulder adduction 2.

1.2.4. Abduction

Raw data → 0°

The equipment and procedure to perform this exercise is exactly the same as for shoulder adduction, only in this one the bench should be placed closer to the anchor point (Figure 7).



Figure 7: Shoulder abduction.

1.3. Elbows

1.3.1. Extension

Projection of the force exerted → **Subtract the effect of the mass from raw data** → -90°

To properly assess the elbow extension force, the following material is needed:

- Two straps.
- An anchor point on the ceiling or on any surface at a certain height.
- A triceps training rope in case you want to perform it bilaterally (Figure 8) or a strap with a handle if you want to perform it unilaterally (Figure 9).



Figure 8: Bilateral elbow extension.

The procedure for performing this test is as follows:

1. A strap is anchored to the anchor point, followed by the force sensor and the training rope or strap with handle.
2. The subject grasps the rope or strap and stands so that the elbow makes a ninety-degree angle and the straps are completely vertical; perpendicular to the ceiling and floor.
3. Because the option "Subtract mass effect from raw data" has been selected. Chronojump will perform a tare.
4. Once the tare is done, the exercise can be performed.



Figure 9: Unilateral elbow extension.

1.3.2. Flexion

Raw data → 90°

The following equipment is needed to properly assess the bending strength of the elbows:

- Bench, stool or any seat on which neither the arms nor the back are supported.
- Two straps, one of which must be padded as in the image (Figure 10), or must have a handle for gripping.
- A floor anchorage.
- An elbow support surface higher than the seat.

The procedure for this test is as follows:

1. The unpadded strap is anchored to the anchor and the seat and support surface are placed close together as shown.
2. The subject sits so that the arm is below ninety degrees to the trunk and grasps the strap so that the elbow is at ninety degrees.
3. If it is necessary to move the support point, the seat, or if it is necessary to tighten or loosen the straps, this is done. In this way the subject will perform the test more comfortably.
4. The exercise can now be performed.



Figure 10: Elbow flexion.

1.5. Trunk

1.5.1. Extension

Raw data → 0°

To properly assess trunk extension strength, the following equipment is needed:

- A bench.
- Two straps, one of which must be padded.
- An anchor point at a certain height that allows the strap to be attached.
- It is advisable to have a stop for the bench and for the feet.

The procedure for this test is as follows:

1. The unpadded strap is attached to the anchor point and the bench is placed perpendicular to the surface where the anchor point is (Figure 11).
2. The subject sits on the bench facing the anchor and the padded strap is placed around the trunk, with the arms on the inside to act as a stop. The straps are tightened or loosened so that they are completely horizontal; parallel to the bench.
3. The exercise can now be performed.
4. If the subject feels that he/she is exerting force with the legs and that this force is altering the result, he/she should extend them more and adjust the strap again until he/she feels that he/she is not really using them.
5. If the subject feels that he/she is exerting force by extending the hips and that this force is altering the result, he/she should grab the bench from the front to avoid doing this hip extension.



Figure 11: Trunk extension.

1.5.2. Flexion

Raw data → 0°

For this exercise the material required is the same as for the previous trunk extension test, but the procedure is somewhat different:

1. All material is placed in the same manner as in the trunk extension test.
2. The subject sits with his/her back to the anchor and, this time, with the trunk slightly tilted backward (Figure 12).
3. The padded strap is placed around the trunk and the exercise can now be performed.
4. For more stability, the legs can be attached to the bench with a rope/strap/rubber.

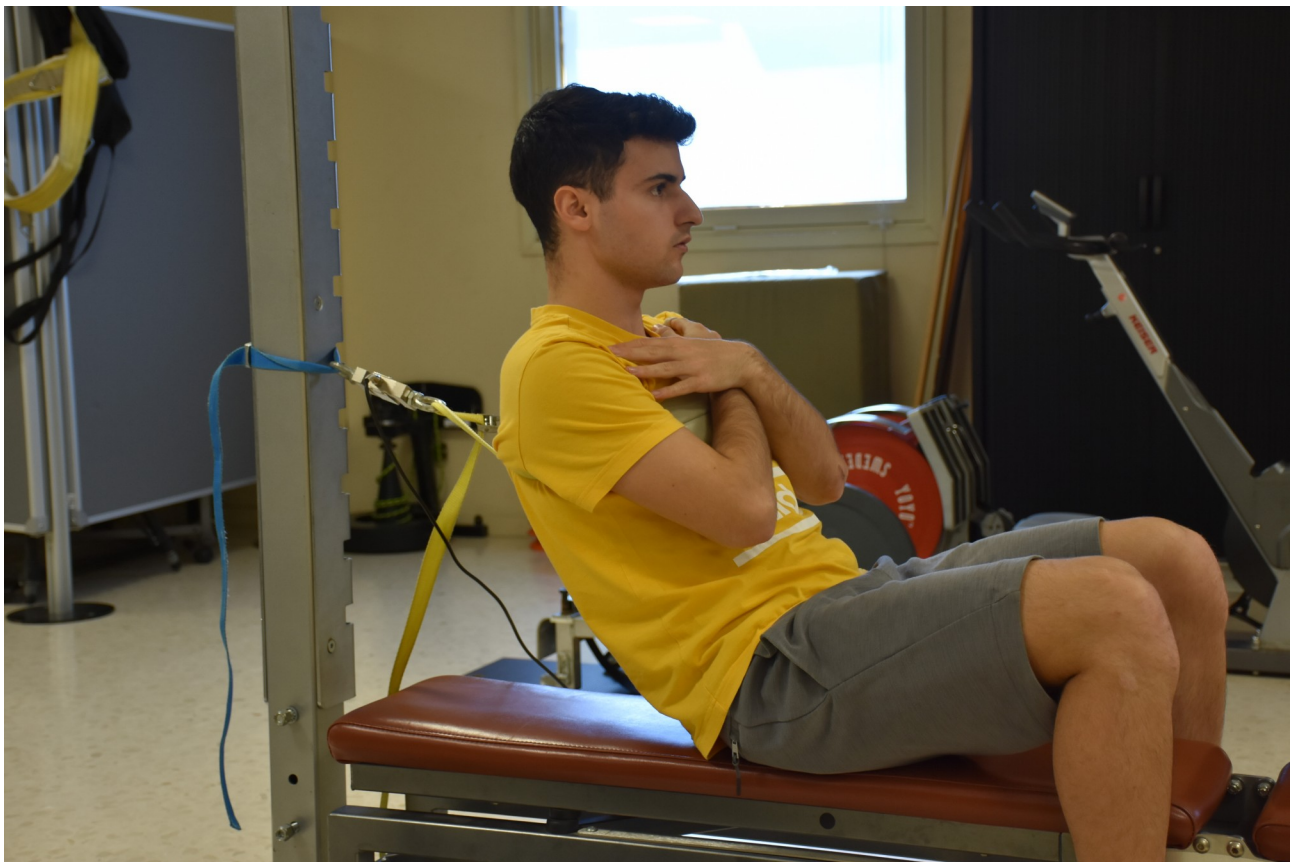


Figure 12: Trunk flexion.

2. Lower trunk

2.1. Hip

2.1.1. Extension

Projection of the force exerted → Subtract the effect of the mass from raw data → -90°

To properly assess hip extension strength, the following equipment is required:

- Two straps, one of which must be padded.
- A bench, stretcher or any surface where you can lie down comfortably.
- An anchor point on the ceiling or any surface at a certain height.

The procedure for this test is as follows:

1. A strap is anchored to the anchor point, followed by the force sensor and the padded strap. The bench or stretcher is also placed under the anchor.
2. The subject lies face up and holds one leg with the padded strap.
3. Because the option "Subtract mass effect from raw data" has been selected. Chronojump will perform a tare.
4. Once the previous steps have been completed, the exercise can be performed (Figures 13 and 14).



Figure 13: Hip extension.



Figure 14: Hip extension 2.

This other exercise can also be done with the same material:

Raw data → 0°

1. A different anchor point is required (Figure 15).
2. The subject lies down with the legs in the opposite direction to where the anchor point is.
3. The subject lifts the leg up to about ninety degrees and holds it with the padded strap.
4. Once the leg is secured, the exercise can be performed. In this case there is no need for a tare, as the force sensor is horizontal.



Figure 15: Second exercise hip extension.

In hip flexion, hip adduction and knee flexion exercises the material is the same.

2.1.2. Flexion

Raw data → 0°

In this exercise the equipment is the same as in the hip extension exercises. The arrangement of the material is the same as in the second hip extension exercise, only that the straps and the sensor have to be at a lower height. The procedure to perform this test is as follows:

1. The subject lies on his back with his feet pointing to the anchor.
2. The strap is placed around one foot in suspension, so that both the knee and hip make a ninety degree angle in flexion (Figure 16).
3. Adjust the straps if necessary and the test can be performed.



Figure 16: Hip flexion.

2.1.3. Adduction

Projection of the force exerted → Subtract the effect of the mass from raw data → -90°

In this exercise the material is the same as in the hip extension exercise. The arrangement of the material is the same as in the first hip extension exercise. The procedure for performing this test is as follows:

1. The athlete lies on his side with his legs under the anchor point.
2. The padded strap is placed around the top foot so that the leg is at about thirty degrees (Figures 17 and 18) to the bench or stretcher.
3. Because the option "Subtract mass effect from raw data" has been selected. Chronojump will perform a tare.
4. Once the tare is done, the exercise can be performed.

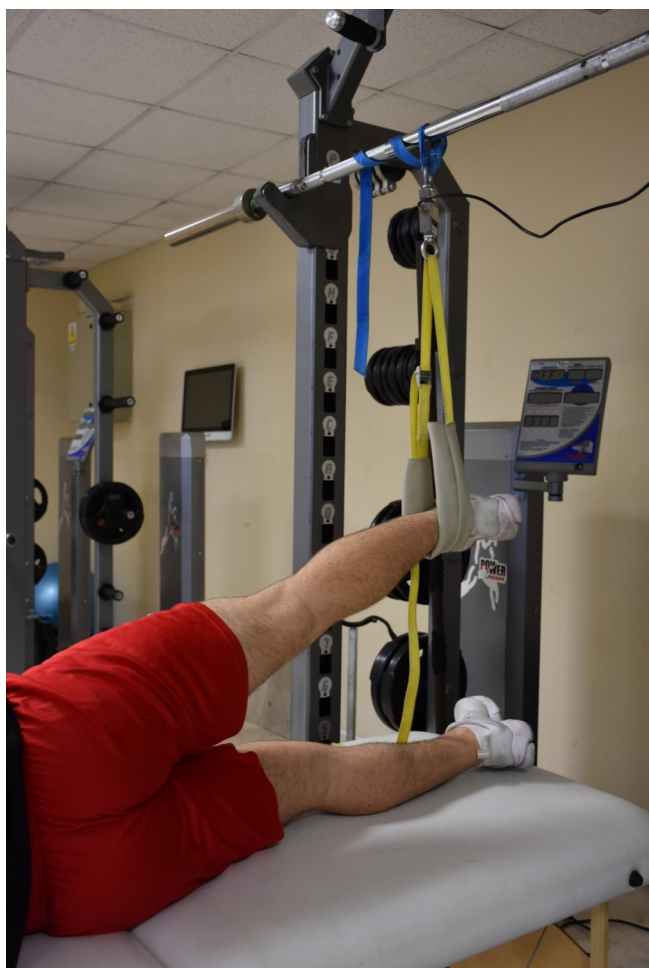


Figure 17: Hip adduction.



Figure 18: Hip adduction 2.

2.1.4. Abduction

Raw data → 0°

To properly assess hip abduction strength, the following equipment is required:

- Two straps, one of which must be padded.
- An anchor point, either on the floor or on another surface, but at a low height, as in the image (Figures 19 and 20).
- A support point to hold on to.

The procedure for this test is as follows:

1. The unpadded strap is anchored and the padded strap is placed around the leg away from the anchor point.
2. The subject opens the leg about thirty degrees with respect to the vertical of the trunk and can now perform the test.
3. In this exercise it is necessary to have a support point to hold on to in order to keep the body straight and still to measure only the force of abduction of the hip.



Figure 19: Hip abduction.



Figure 20: Hip abduction 2.

2.2. Knee

2.2.1. Extension

Raw data → 0°

To properly assess hip abduction strength, the following equipment is required:

- Two straps, one of which must be padded.
- A seat with legs to anchor the unpadded strap.

The procedure for this test is as follows:

1. The unpadded strap is anchored to one leg of the seat.
2. The subject sits on the edge of the seat and can now perform the test.
3. It is recommended that the knee and hip be in a flexed position of about ninety degrees. It is also advisable to use the arms to hold on to the seat for stability (Figure 21).



Figure 21: Knee extension.

2.2.2. Flexion

Raw data → 0°

In this exercise the equipment required is the same as in the hip extension exercise. The arrangement of the material is the same as in the second hip extension exercise, only that the straps and the sensor have to be at a lower height.

The procedure for performing this test is as follows:

1. The subject lies face down with the legs pointing towards the anchor point.
2. One knee is flexed and the padded strap is placed around the ankle.
3. The knee should make about ninety degrees with the rest of the body and the field or stretcher, while the straps and sensor should be completely horizontal (Figure 22).
4. If necessary, adjust the straps so that the starting position is correct and the test can be performed.



Figure 22: Knee flexion.

Another exercise that can be performed to assess knee flexion strength is the following.

Projection of the force exerted → Subtract the effect of the mass from raw data → -90°

In this exercise the material is the same as in the hip extension exercise. The arrangement of the material is the same as in the first hip extension exercise. The procedure for performing this test is as follows:

1. The subject lies face up with the legs just below the anchor point.
2. He lifts one leg and places the strap around the ankle.
3. It is important that both the knee and hip are in ninety degrees of flexion. If necessary, adjust the straps or the position of the subject to be in the correct position (Figures 23 and 24).
4. Once well positioned, the exercise can be performed.



Figure 23: Segundo ejercicio flexión de rodilla.

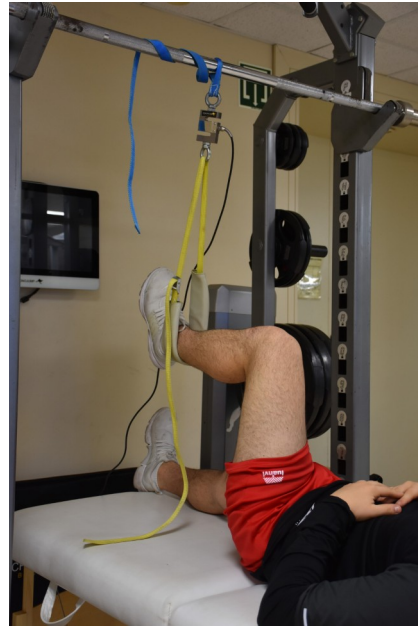


Figure 24: Segundo ejercicio flexión de rodilla 2.

2.3. Ankle

2.3.1. Plantar flexion

Projection of the force exerted → Add the effect of the mass to raw data (bar weight) → 90°

To properly assess the plantar flexion strength of the ankle, the following equipment is required:

- Two straps.
- A seat.
- A floor anchor.
- A light bar, such as the one used in the shoulder flexion exercise.

The procedure to perform this test is as follows:

1. One strap is anchored to the floor anchor, followed by the force sensor and the other strap, which is attached to the light bar.
2. The subject sits on the seat; it is recommended that the hips and knees make ninety degrees of flexion, and it is also advisable to use the arms to hold on to the seat for greater stability (Figure 25).
3. The heel should not be resting on the floor, but slightly elevated (Figure 26).
4. The bar is placed above the knees, not too close to the edge so that it does not slip during the exercise, and so that the force sensor remains vertical.
5. Tighten the straps so that they are well tensioned and the exercise can be performed.

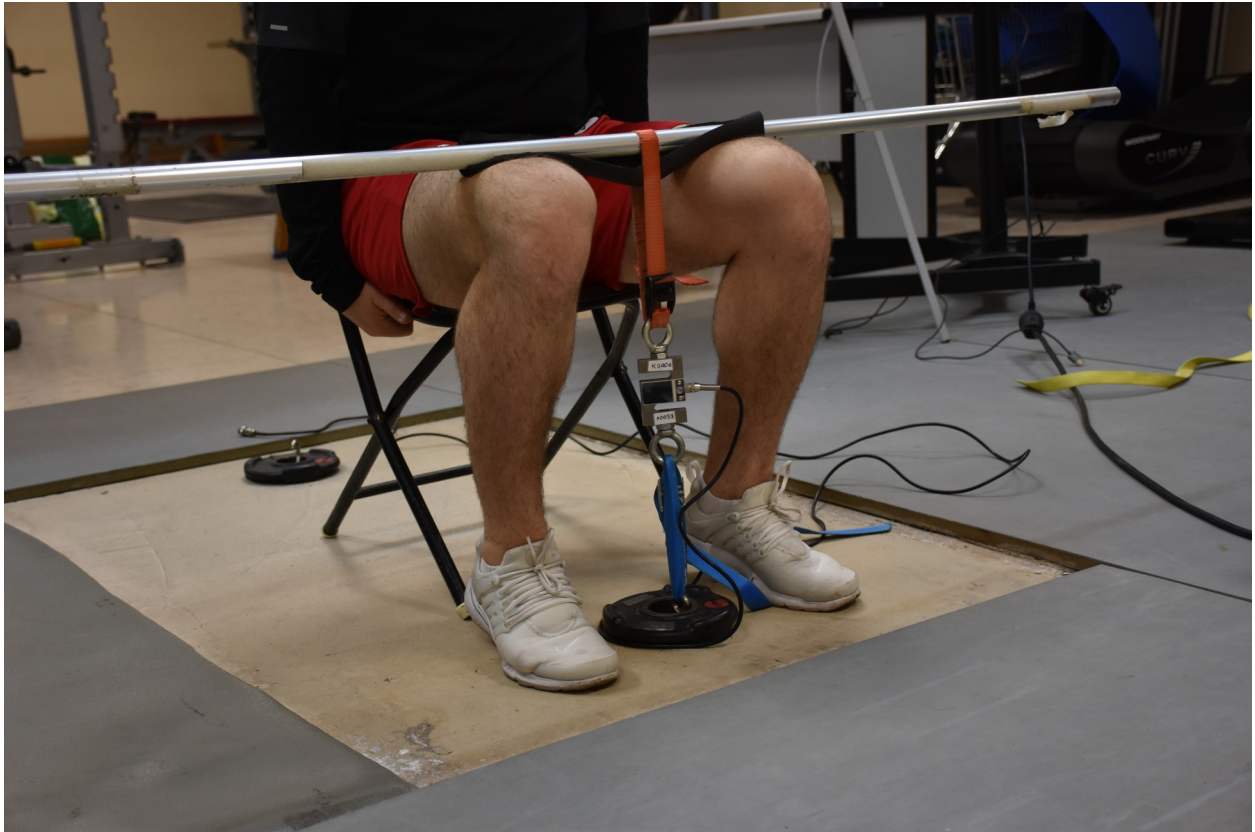


Figure 25: Ankle plantar flexion.



Figure 26: Ankle plantar flexion 2.

This exercise can also be performed standing without a seat.

Projection of the force exerted → Add the effect of the mass to raw data (body weight) → 90°

Instead of holding the bar on the knees, the bar is grasped with the hands (Figures 27 and 28).

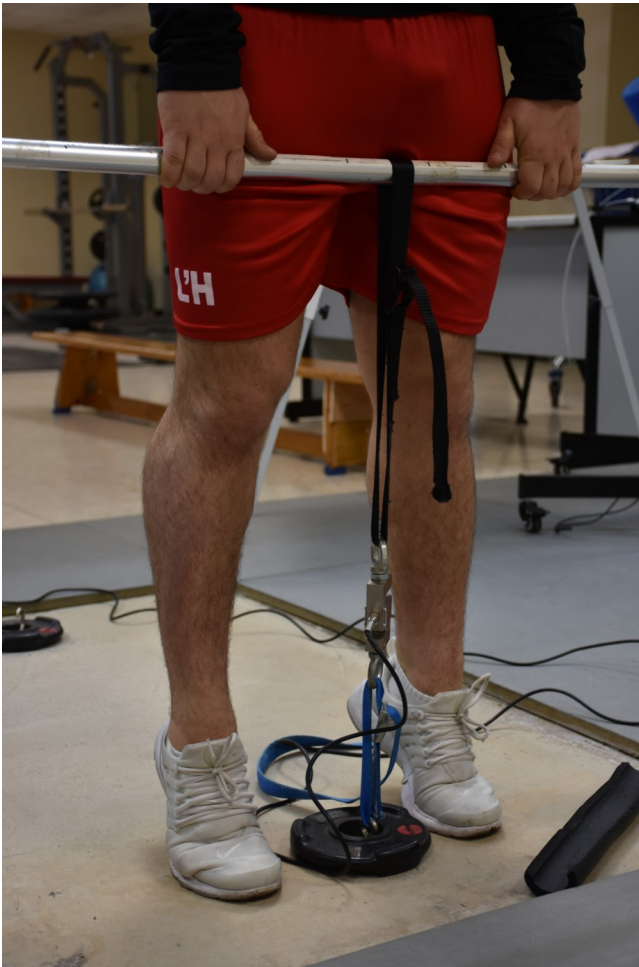


Figure 27: Ankle plantar flexion (standing).

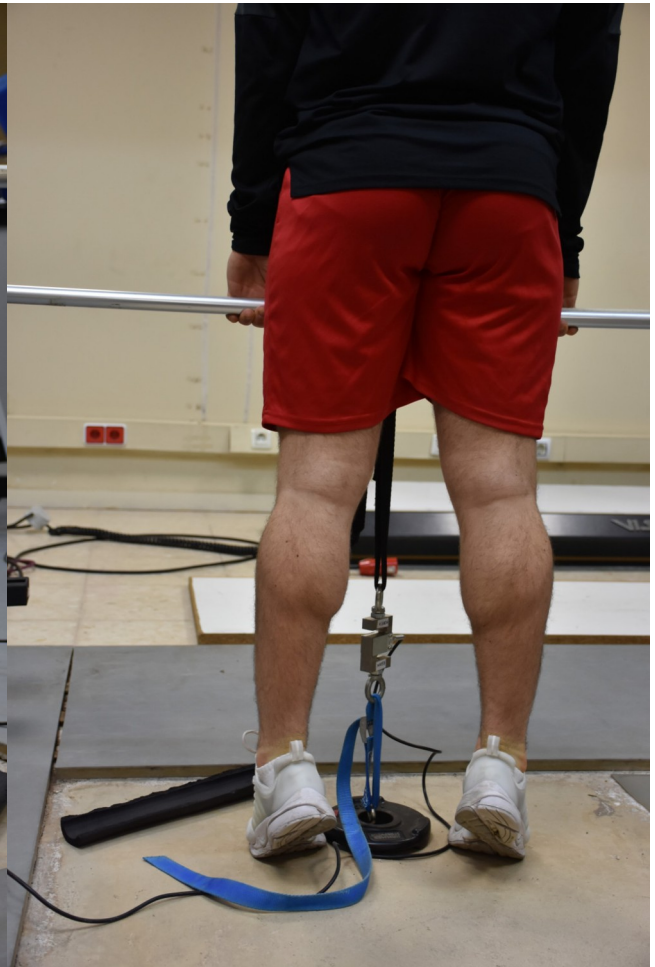


Figure 28: Ankle plantar flexion (standing).

2.3.2. Dorsal flexion

Raw data → 30°

To properly assess the dorsal flexion strength of the ankle, the following equipment is required:

- Two straps, one of which has to be padded.
- A seat.
- A floor anchor.

The procedure for this test is as follows:

1. The unpadded strap is anchored to the floor anchor.
2. The subject sits on the seat and the padded strap is placed around the metatarsal of the foot.
3. The subject leaves the leg in suspension without reaching maximum knee extension so that the straps are taut and aligned with the leg (Figure 29).
4. It is recommended to use the arms to hold on to the seat for greater stability.
5. Once the previous steps have been completed, the test can be performed.



Figure 29: Dorsal ankle flexion.

3. Compounds

3.1. Mid-thigh pull

Projection of the force exerted → Add the effect of the mass to raw data (body weight) → 90°

To properly assess the strength of the mid-thigh pull, the following material is needed:

- Two straps.
- An Olympic bar.
- A floor anchor.

The procedure to perform this test is as follows:

1. One strap is anchored to the floor anchor, followed by the force sensor and the other strap, which holds the Olympic bar.
2. The subject grasps the bar and stands with the knees semi-flexed and the hips also slightly flexed so that the trunk is straight, but slightly bent forward (Figures 30 and 31).
3. Once the subject is well positioned, the straps are tightened so that they are well tensioned and the exercise can be performed.



Figure 30: Mid-thigh pull.

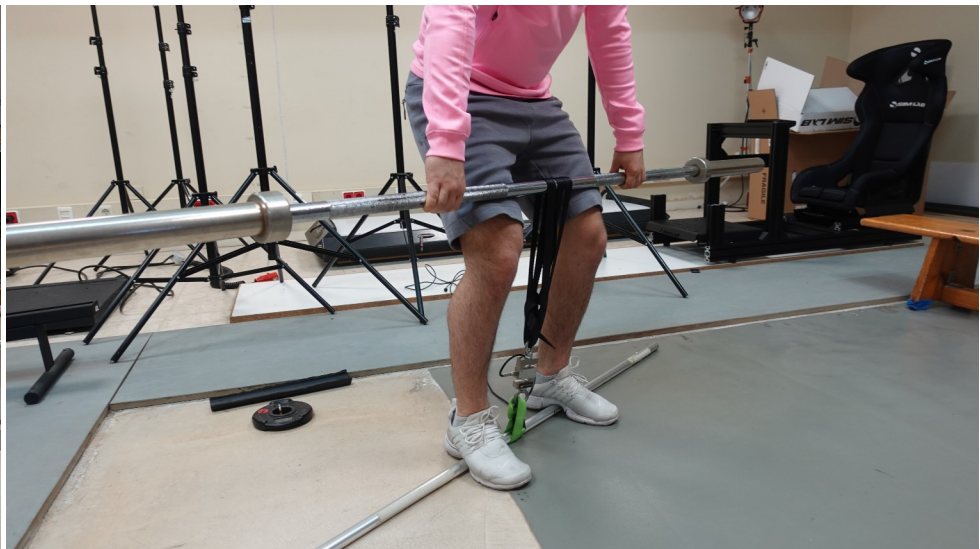


Figure 31: Mid-thigh pull 2.

3.2. Row

Raw data → 0°

To properly assess the strength in the rowing exercise, the following material is needed:

- Three straps, two of which must have handles.
- A carabiner.
- A bench.
- An anchor point at a certain height that allows the strap to be attached.
- A stop for the foot or that the support surface is not slippery so that the subject remains fixed when performing the test.

The procedure for this test is as follows:

1. The strap is anchored to the anchor point, followed by the force sensor, the carabiner, and the two straps with handle.
2. The subject sits on the bench with his trunk straight and slightly away from the straps, as he will hold them with his arms fully extended.
3. The subject grasps the straps, which must be taut and aligned with the subject's arms.
4. Make sure that the feet will not move during the test by placing the foot stops as shown in the image (Figure 32), or by checking that the support surface is not slippery.
5. Once well positioned, the subject can now perform the test.

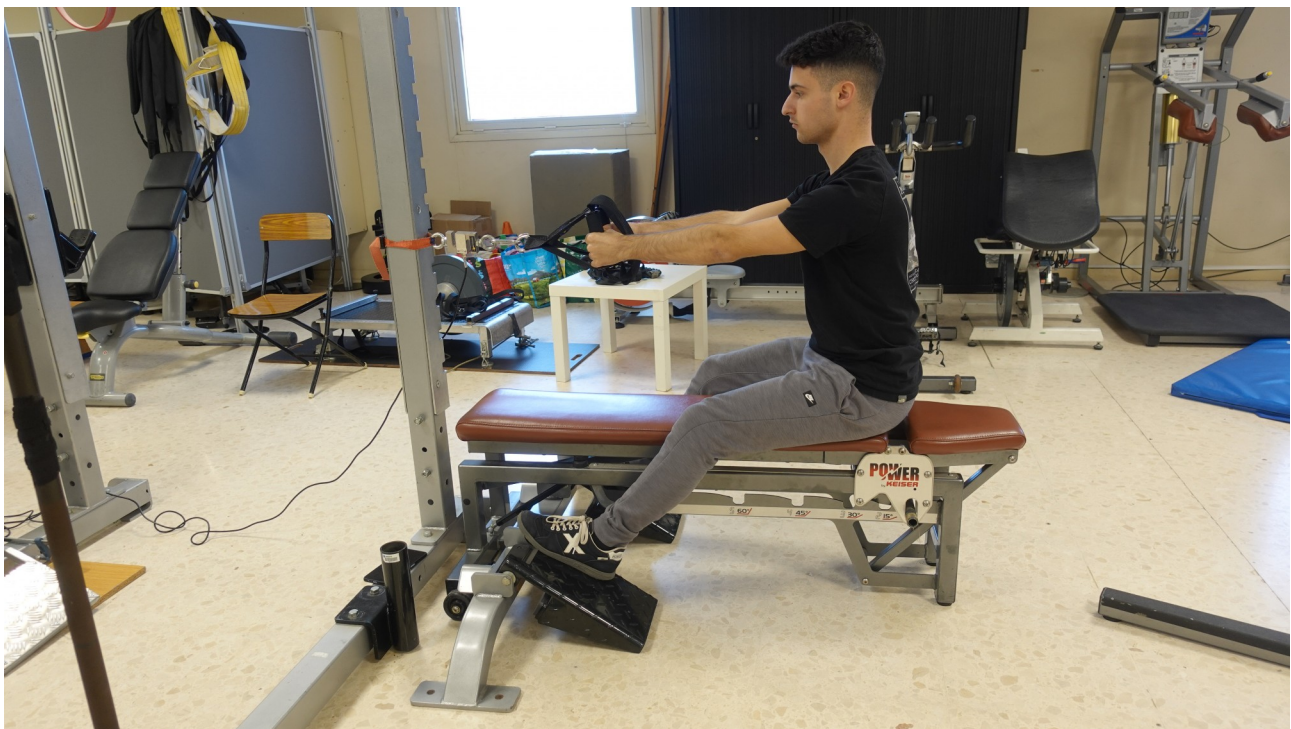


Figure 32: Row.